

DDR4-UDIMM routing guide

16 GB DDR4-3200 UDIMM, Raw Card A3. Which signal is routed on which layer, by component designator. Tables generated from DDR4-UDIMM — 16 GB DDR4.PcbDoc; rules from JESD21-C 4.20.26 (main spec Tables 10-14 and Annex A).

1. Layer stack

Layer	Name	Type	References	What is routed here
L1	Top Layer 1	signal, microstrip	L2 (plane, 70 um)	DQ / DQS / DM escapes, finger stubs, short address hops
L2	L2_PWR_GND	plane	-	reference for L1 and L3; GND under the data groups
L3	L3_DQ_ADDR	signal, stripline	L2 (plane, 80 um)	DQ / DQS / DM DRAM side; address permitted
L4	L4_VDD	plane	-	reference for L5; VDD
L5	L5_ADDR	signal, stripline	L4 (VDD, 80 um)	address and command only
L6	L6_ADDR_CK	signal, stripline	L7 (plane, 80 um)	address, command and the clock pairs
L7	L7_PWR_GND	plane	-	reference for L6 and L8
L8	Bottom Layer 1	signal, microstrip	L7 (plane, 70 um)	DQ / DQS / DM DRAM side, clocks

Never route on L2, L4 or L7. They are solid planes and they are the return path for everything else; a track on a plane cuts that return path.

2. Layer assignment rules (enforced by DRC)

Signal group	Allowed layers	Altium rule	Why
DQ / DQS / DM	L1, L3, L8 only	RoutingLayers_DATA	L5 and L6 reference VDD; data return current needs GND (Table 12)
CK0_T / CK0_C	L1, L6, L8 only	RoutingLayers_CK	clock references VDD or a plane, per Table 12
Address, control, RESET_n, ALERT_n	L1, L3, L5, L6, L8	RoutingLayers_ADDR	address references VDD
Everything else	L1, L3, L5, L6, L8	RoutingLayers	L2, L4 and L7 are planes - never route on them

3. Layer by layer

L1 - Top Layer 1

signal (microstrip, references L2)

Signal group	Components	From - to	Width / impedance
Data, all 8 bytes	R2-R12, R14-R24, ... R86-R96 (88 resistors)	J1 finger to resistor pad 2, and resistor pad 1 to the DRAM fan-out via	0.10 mm, 50 ohm
Address and command	R97-R126 to J1 and the DRAMs	finger stubs and short hops	0.075 mm, 55 ohm
ZQ	R1, R13, R25, R37, R49, R61, R73, R85	DRAM ZQ ball to 240 ohm to GND	0.10 mm
ALERT_n	R102	47 ohm pull-up to VDD, before the first DRAM (U1)	0.075 mm

L3 - L3_DQ_ADDR

signal (stripline, references L2 = GND)

Signal group	Components	From - to	Width / impedance
Data, DRAM side	NetR2_1 ... NetR96_1	resistor pad 1 to the DRAM ball, via L1	0.10 mm, 50 ohm
Address and command	any of A0-A16, BA, BG, ACT, CE, CS, ODT, PAR, WE	permitted but keep it clear of the byte lanes	0.075 mm, 55 ohm

L5 - L5_ADDR

signal (stripline, references L4 = VDD)

Signal group	Components	From - to	Width / impedance
Address	A0-A13, A10_AP, A12_BC, A15_CAS, A16_RAS	J1 to U1 -> U2 -> ... -> U8 fly-by	0.075 mm, 55 ohm
Bank / command	BA0, BA1, BG0, BG1, ACT, WE, PAR	same fly-by	0.075 mm, 55 ohm
Control	CKE, CS, ODT, RESET_n	same fly-by	0.075 mm, 55 ohm

L6 - L6_ADDR_CK

signal (stripline, references L7)

Signal group	Components	From - to	Width / impedance
Clock pair	CK0_T / CK0_C	J1 to every DRAM, fly-by, terminated at R115 / R117 and C56	0.075 / 0.10 mm gap, 93 ohm differential
Address and command overflow	whatever does not fit on L5	same fly-by	0.075 mm, 55 ohm

L8 - Bottom Layer 1

signal (microstrip, references L7)

Signal group	Components	From - to	Width / impedance
Data, DRAM side	NetR2_1 ... NetR96_1	second choice when L3 is congested	0.10 mm, 50 ohm
Clocks	CK0_T / CK0_C	permitted	93 ohm differential

4. Data byte lanes, by designator

Every data line is J1 finger -> Rn pad 2 -> Rn pad 1 (15 ohm) -> DRAM ball. Route pad 1 to the ball on L1 + L3 (or L1 + L8). The target is the byte's longest line; each line must land within 1.0 mm of it. **Route the whole byte before tuning any of it:** the target below is only the longest line routed *so far*, so in a byte that is still mostly unrouted it will rise. A DRAM-side length of about 0.57 mm means nothing is routed yet - that is just the ball-to-via stub.

Byte 0 - U1 (target 19.56 mm, window 18.56 to 20.56)

Resistor	Signal	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R2	DQ0	U1-C2	L1 + L3 or L1 + L8	2.95	9.43	13.38	+6.19 mm
R3	DQ1	U1-B7	L1 + L3 or L1 + L8	5.45	4.00	10.45	+9.11 mm
R4	DQ2	U1-D3	L1 + L3 or L1 + L8	2.95	8.27	12.22	+7.34 mm
R5	DQ3	U1-D7	L1 + L3 or L1 + L8	5.45	11.21	17.66	+1.91 mm
R6	DQ4	U1-D2	L1 + L3 or L1 + L8	2.95	11.02	14.97	+4.59 mm
R7	DQ6	U1-E3	L1 + L3 or L1 + L8	2.95	12.06	16.01	+3.56 mm
R8	DQ7	U1-E7	L1 + L3 or L1 + L8	5.45	13.11	19.56	in window
R9	DQS0_T	U1-C3	L1 + L3 or L1 + L8	5.54	7.01	13.55	+6.01 mm
R10	DQS0_C	U1-B3	L1 + L3 or L1 + L8	5.54	6.68	13.22	+6.34 mm
R11	DM0	U1-A7	L1 + L3 or L1 + L8	2.95	6.00	9.95	+9.61 mm
R12	DQ5	U1-D8	L1 + L3 or L1 + L8	5.45	5.92	12.37	+7.20 mm

Byte 1 - U2 (target 14.78 mm, window 13.78 to 15.78)

Resistor	Signal	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R14	DQ8	U2-C2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R15	DQ9	U2-B7	L1 + L3 or L1 + L8	5.47	0.74	7.21	route it
R16	DQ10	U2-D3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R17	DQ11	U2-D7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R18	DQ12	U2-D2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R19	DQ14	U2-E3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R20	DQ15	U2-E7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R21	DQS1_T	U2-C3	L1 + L3 or L1 + L8	5.54	8.24	14.78	in window

R22	DQS1_C	U2-B3	L1 + L3 or L1 + L8	5.54	8.24	14.78	in window
R23	DM1	U2-A7	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R24	DQ13	U2-D8	L1 + L3 or L1 + L8	5.45	7.95	14.40	in window

Byte 2 - U3 (target 16.43 mm, window 15.43 to 17.43)

Resistor	Signal	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R26	DQ16	U3-C2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R27	DQ17	U3-B7	L1 + L3 or L1 + L8	5.47	0.57	7.04	route it
R28	DQ18	U3-D3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R29	DQ19	U3-D7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R30	DQ20	U3-D2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R31	DQ22	U3-E3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R32	DQ23	U3-E7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R33	DQS2_T	U3-C3	L1 + L3 or L1 + L8	5.56	8.12	14.69	+1.74 mm
R34	DQS2_C	U3-B3	L1 + L3 or L1 + L8	5.56	9.87	16.43	in window
R35	DM2	U3-A7	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R36	DQ21	U3-D8	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it

Byte 3 - U4 (target 17.33 mm, window 16.33 to 18.33)

Resistor	Signal	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R38	DQ24	U4-C2	L1 + L3 or L1 + L8	3.01	0.57	4.58	route it
R39	DQ25	U4-B7	L1 + L3 or L1 + L8	5.53	0.57	7.10	route it
R40	DQ26	U4-D3	L1 + L3 or L1 + L8	3.16	0.57	4.72	route it
R41	DQ27	U4-D7	L1 + L3 or L1 + L8	5.68	0.57	7.24	route it
R42	DQ28	U4-D2	L1 + L3 or L1 + L8	2.97	0.57	4.54	route it
R43	DQ30	U4-E3	L1 + L3 or L1 + L8	3.12	0.57	4.68	route it
R44	DQ31	U4-E7	L1 + L3 or L1 + L8	5.64	0.57	7.20	route it
R45	DQS3_T	U4-C3	L1 + L3 or L1 + L8	5.71	10.62	17.33	in window
R46	DQS3_C	U4-B3	L1 + L3 or L1 + L8	5.67	10.62	17.29	in window
R47	DM3	U4-A7	L1 + L3 or L1 + L8	3.05	0.57	4.62	route it
R48	DQ29	U4-D8	L1 + L3 or L1 + L8	5.49	0.57	7.06	route it

Byte 4 - U5 (target 14.73 mm, window 13.73 to 15.73)

Resistor	Signal	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R50	DQ32	U5-C2	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R51	DQ33	U5-B7	L1 + L3 or L1 + L8	5.70	0.57	7.27	route it
R52	DQ34	U5-D3	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R53	DQ35	U5-D7	L1 + L3 or L1 + L8	5.70	0.57	7.27	route it
R54	DQ36	U5-D2	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R55	DQ38	U5-E3	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R56	DQ39	U5-E7	L1 + L3 or L1 + L8	5.70	0.57	7.27	route it
R57	DQS4_T	U5-C3	L1 + L3 or L1 + L8	5.79	7.91	14.71	in window
R58	DQS4_C	U5-B3	L1 + L3 or L1 + L8	5.81	7.91	14.73	in window
R59	DM4	U5-A7	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R60	DQ37	U5-D8	L1 + L3 or L1 + L8	5.70	0.57	7.27	route it

Byte 5 - U6 (target 12.65 mm, window 11.65 to 13.65)

Resistor	Signal	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R62	DQ40	U6-C2	L1 + L3 or L1 + L8	3.14	0.57	4.70	route it
R63	DQ41	U6-B7	L1 + L3 or L1 + L8	5.62	0.57	7.18	route it
R64	DQ42	U6-D3	L1 + L3 or L1 + L8	2.99	0.57	4.56	route it
R65	DQ43	U6-D7	L1 + L3 or L1 + L8	5.47	0.57	7.04	route it
R66	DQ44	U6-D2	L1 + L3 or L1 + L8	3.18	0.57	4.74	route it
R67	DQ46	U6-E3	L1 + L3 or L1 + L8	3.03	0.57	4.60	route it
R68	DQ47	U6-E7	L1 + L3 or L1 + L8	5.51	0.57	7.08	route it
R69	DQS5_T	U6-C3	L1 + L3 or L1 + L8	5.65	5.96	12.61	in window
R70	DQS5_C	U6-B3	L1 + L3 or L1 + L8	5.69	5.96	12.65	in window
R71	DM5	U6-A7	L1 + L3 or L1 + L8	3.10	0.57	4.66	route it
R72	DQ45	U6-D8	L1 + L3 or L1 + L8	5.66	0.57	7.22	route it

Byte 6 - U7 (target 12.04 mm, window 11.04 to 13.04)

Resistor	Signal	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R74	DQ48	U7-C2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R75	DQ49	U7-B7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R76	DQ50	U7-D3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R77	DQ51	U7-D7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R78	DQ52	U7-D2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R79	DQ54	U7-E3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R80	DQ55	U7-E7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R81	DQS6_T	U7-C3	L1 + L3 or L1 + L8	5.54	5.50	12.04	in window
R82	DQS6_C	U7-B3	L1 + L3 or L1 + L8	5.54	5.17	11.71	in window
R83	DM6	U7-A7	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R84	DQ53	U7-D8	L1 + L3 or L1 + L8	5.47	0.57	7.04	route it

Byte 7 - U8 (target 13.02 mm, window 12.02 to 14.02)

Resistor	Signal	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R86	DQ56	U8-C2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R87	DQ57	U8-B7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R88	DQ58	U8-D3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R89	DQ59	U8-D7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R90	DQ60	U8-D2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R91	DQ62	U8-E3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R92	DQ63	U8-E7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R93	DQS7_T	U8-C3	L1 + L3 or L1 + L8	5.56	6.45	13.02	in window
R94	DQS7_C	U8-B3	L1 + L3 or L1 + L8	5.54	6.45	12.99	in window
R95	DM7	U8-A7	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R96	DQ61	U8-D8	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it

5. Address, command and control

Fly-by from J1 through U1 to U8 and then into the 39 ohm termination to VTT. Route on **L5** and **L6**, which are empty and reference VDD as Table 12 requires. Width 0.075 mm (55 ohm). Within 1.0 mm of each other and within CK +/- 0.5 mm.

Resistor	Signal	Function	Layers	Status
R97	A0	39 ohm to VTT	L5 / L6	done
R98	BG1	39 ohm to VTT	L5 / L6	done
R99	BG0	39 ohm to VTT	L5 / L6	done
R100	A1	39 ohm to VTT	L5 / L6	done
R101	CS	39 ohm to VTT	L5 / L6	done
R103	A2	39 ohm to VTT	L5 / L6	done
R104	CKE	39 ohm to VTT	L5 / L6	done
R105	A3	39 ohm to VTT	L5 / L6	done
R107	ODT	39 ohm to VTT	L5 / L6	done
R108	A4	39 ohm to VTT	L5 / L6	done
R109	WE	39 ohm to VTT	L5 / L6	done
R110	A5	39 ohm to VTT	L5 / L6	done
R111	A15_CAS	39 ohm to VTT	L5 / L6	done
R112	A6	39 ohm to VTT	L5 / L6	done
R113	A16_RAS	39 ohm to VTT	L5 / L6	done
R114	A7	39 ohm to VTT	L5 / L6	done
R116	A8	39 ohm to VTT	L5 / L6	done
R118	BA0	39 ohm to VTT	L5 / L6	done
R119	A9	39 ohm to VTT	L5 / L6	done
R120	BA1	39 ohm to VTT	L5 / L6	done
R121	A10_AP	39 ohm to VTT	L5 / L6	done
R122	A11	39 ohm to VTT	L5 / L6	done
R123	A12_BC	39 ohm to VTT	L5 / L6	done
R124	A13	39 ohm to VTT	L5 / L6	done
R125	ACT	39 ohm to VTT	L5 / L6	done
R126	PAR	39 ohm to VTT	L5 / L6	done

6. Clocks, ZQ, ALERT_n and SPD

Components	Signals	Function	Layers	Notes
R115 / R117 + C56	CK0_T / CK0_C	39 ohm each plus 0.01 uF to VDD, after the last DRAM	L1, L6, L8	differential pair, 93 ohm, gap 0.10 mm
R106	CK1_T / CK1_C	75 ohm across the unused clock pair	L1, L6, L8	terminate at the connector end
R1, R13, R25, R37, R49, R61, R73, R85	ZQ of U1-U8	240 ohm 1% to GND	L1	keep it short, straight to the ball
R102	ALERT_n	47 ohm pull-up to VDD	L1, L3, L5, L6, L8	must sit BEFORE the first DRAM (Annex A 6.3.7)
U9	SA0, SA1, SA2, SCL, SDA	SPD EEPROM (34AA04)	any signal layer	low speed, no length rules

7. Differential pairs

Pair	Positive	Negative	Where	Layers
CK0	CK0_T	CK0_C	connector side	L1, L6, L8
CK1	CK1_T	CK1_C	connector side	L1, L6, L8

DQS0	DQS0_T	DQS0_C	connector side	L1, L3, L8
DQS0_DRAM	NetR9_1	NetR10_1	DRAM side	L1, L3, L8
DQS1	DQS1_T	DQS1_C	connector side	L1, L3, L8
DQS1_DRAM	NetR21_1	NetR22_1	DRAM side	L1, L3, L8
DQS2	DQS2_T	DQS2_C	connector side	L1, L3, L8
DQS2_DRAM	NetR33_1	NetR34_1	DRAM side	L1, L3, L8
DQS3	DQS3_T	DQS3_C	connector side	L1, L3, L8
DQS3_DRAM	NetR45_1	NetR46_1	DRAM side	L1, L3, L8
DQS4	DQS4_T	DQS4_C	connector side	L1, L3, L8
DQS4_DRAM	NetR57_1	NetR58_1	DRAM side	L1, L3, L8
DQS5	DQS5_T	DQS5_C	connector side	L1, L3, L8
DQS5_DRAM	NetR69_1	NetR70_1	DRAM side	L1, L3, L8
DQS6	DQS6_T	DQS6_C	connector side	L1, L3, L8
DQS6_DRAM	NetR81_1	NetR82_1	DRAM side	L1, L3, L8
DQS7	DQS7_T	DQS7_C	connector side	L1, L3, L8
DQS7_DRAM	NetR93_1	NetR94_1	DRAM side	L1, L3, L8

Use **Route > Interactive Differential Pair Routing** (U, P) for these, never two single tracks. Halves matched within 0.1 mm; the pair as a whole is matched to its byte.

8. Special cases and things that catch people out

Differential pairs

18 pairs: CK0, CK1 and DQS0-DQS7 on the connector side, plus DQS0_DRAM-DQS7_DRAM between the 15 ohm resistors and the DRAMs. Route with **Route > Interactive Differential Pair Routing** (U, P) so both wires are drawn together and stay phase matched. Gap 0.10 mm, halves matched within 0.1 mm (main spec Table 11, rule MatchedLengths_DDR_CLOCKS_STROBES).

Byte-lane length matching

Each DQ and DM must sit within DQS +/- 1.0 mm over the whole finger-to-ball path, and the byte total must be 12.0-32.0 mm compensated (Table 11). The longest line in the byte is the reference: adding length to it lengthens every other line in that byte, so never tune the longest one.

xSignals

Once a byte is routed, **Design > xSignals > Create xSignals** with source J1 and destination Un traces the finger -> 15 ohm -> ball path as one object (4 nodes). Put the 11 into an xSignal class BYTEn_FULL and point a Matched Lengths rule at it with tolerance 1 mm. It fails on unrouted nets, so route first, then create.

Address fly-by

J1 -> U1 -> U2 -> U3 -> U4 -> U5 -> U6 -> U7 -> U8, then the 39 ohm termination to VTT. Address and control within 1.0 mm of each other and within CK +/- 0.5 mm; CK first-to-last DRAM at most 153 mm; the stub from the fly-by to each DRAM ball (TL2) at most 3.0 mm (Annex A).

Velocity compensation

All length rules are on stripline-equivalent lengths: divide any microstrip (L1 or L8) length by 1.1 before comparing (main spec 6.3.4). tools/check_lengths.py reports both.

Plane referencing

Table 12: DQ, DQS and DM reference GND; address, command, clock and VREFCA reference VDD. No signal may cross a plane split - check L2, L4 and L7 under every route.

Vias and fan-out

All vias are through-hole, 0.40 mm pad / 0.20 mm drill (0.45 / 0.25 for power). Routing space per Table 14: via-to-track 0.20 mm, large via 0.25 mm, track-to-pad 0.125 mm, track-to-shape 0.20 mm. All vias are tented (rule SolderMask_TentedVias).

Serpentine style

Accordion, Rounded, amplitude up to 1.0 mm in the corridor down the middle of a DRAM and 0.4 mm near the balls, spacing 0.3 mm = 3 x track width. Meander space: the corridor between ball columns 3 and 7 of each DRAM (about 3.2 x 9.6 mm) and the band just above the resistor rows.

9. What is left to route

Byte	DRAM side still to route	Routed but needs length tuning
Byte 0 (U1)	-	R2-R7, R9-R12
Byte 1 (U2)	R14-R20, R23	-
Byte 2 (U3)	R26-R32, R35-R36	R33
Byte 3 (U4)	R38-R44, R47-R48	-
Byte 4 (U5)	R50-R56, R59-R60	-
Byte 5 (U6)	R62-R68, R71-R72	-
Byte 6 (U7)	R74-R80, R83-R84	-
Byte 7 (U8)	R86-R92, R95-R96	-

Address and command: the fly-by from the first DRAM onwards is not routed yet - only the connector stubs exist. Plan it on L5 and L6. Clocks CK1_T / CK1_C are unrouted (terminate at the connector). SPD nets are unrouted.